

Adrian Alan Pol

Switzerland adrianalan.pol@gmail.com +41762791887 adrianalan.me [LinkedIn](#) [Github](#) [Publications](#)

Machine Learning Scientist and Software Engineer with expertise in large language models, anomaly detection, and inference compute optimization for edge devices. Published in leading journals and conferences such as *Nature*, ICLR, NeurIPS, and EMNLP. Experienced in developing systems for both enterprise and scientific applications, including key contributions to a \$500M particle physics experiment at the Large Hadron Collider. Skilled in cross-departmental collaboration, technical leadership, and communicating complex concepts to non-technical stakeholders.

WORK EXPERIENCE

- Thomson Reuters**
 - Applied Machine Learning Scientist* *from Jan 2024*
 - Fine-tuned an ensemble of BERT models on 100M text samples, reducing manual review time by 80%.
 - Compared 15 methods for assessing LLM response hallucinations in legal contexts, improving model reliability.
 - Implemented a real-time regulatory change notification system using RAG and integrating external LLM APIs.
- Princeton University**
 - Machine Learning Postdoctoral Research Associate* *Feb 2022 - Dec 2023*
 - Technical lead on an anomaly detection project aimed at searching for new physics at the Large Hadron Collider.
 - Enhanced sensitivity to novelties by 3x over existing physics baselines using autoencoders and distributed training.
 - Optimized algorithm's inference via quantization, pruning, and distillation, achieving sub-100 ns latency on FPGAs.
 - Taught machine learning at workshops and summer schools to over 200 students, including Nvidia DLI.
- CERN (European Organization for Nuclear Research)**
 - Machine Learning Senior Fellow* *Oct 2019 - Jan 2022*
 - Developed a multimodal particle detection model using the SSD framework, achieving 85% average precision score.
 - Accelerated model inference with TensorRT, achieving 8x speedup and enabling real-time detection on a single GPU.
 - Collaborated with industry partners to prototype AI solutions addressing future challenges at CERN.
 - Co-authored the book *Artificial Intelligence for High Energy Physics* alongside the 20 foremost experts in the field.
 - Machine Learning Doctoral Student* *Oct 2016 - Sep 2019*
 - Engineered ETL pipelines and trained XGBoost to classify high-dimensional physics data (2800+ features).
 - Integrated the first convolutional neural network in the CMS experiment's production code, still in use since 2018.
 - Deployed a service that improved hardware failure detection recall from 26% to 98%, while halving detection delays.
 - Built active learning, retraining, and interpretability APIs using MERN stack, reducing deployment cycle to 1 hour.
 - Software Engineering Intern* *Oct 2014 - Nov 2015*
 - Developed an interactive monitoring tool using Angular, Elasticsearch, and d3.js, saving 6000 person-hours annually.
- Mendix (Siemens)**
 - Software Engineer* *Mar 2016 - Aug 2016*
 - Optimized web and mobile app performance, reducing load times by 60% and introducing a PIN-based login system.
- IBM**
 - Software Engineering Intern* *Jun 2014 - Sep 2014*
 - Prototyped a full-stack mobile app for movie recommendations using content-based and collaborative filtering.

PROJECTS

- United Nations, International Trade Centre**
 - Machine Learning Consultant* *Sep 2024 - Nov 2024*
 - Developed an anomaly detection system to improve fraud detection and revenue collection in customs declarations.
- Codesmith**
 - Data Science and Machine Learning Fellowship* *2023*
 - Created a negotiation assistant using self-instruct to fine-tune LLaMA 7B with negotiation prompts and cured compensation data. Reduced training time to hours by applying LoRA and distributed fine-tuning on A100 GPUs.

EDUCATION

- **University of Paris-Saclay** 2016 - 2020
PhD in Computer Science. Thesis on Machine Learning Anomaly Detection: available here.
- **Technical University of Denmark (exchange at National University of Singapore)** 2012 - 2016
MSc in Telecommunications Engineering. Thesis on Routing Optimization: available here.
- **Lodz University of Technology (exchange at Polytechnic University of Turin)** 2008 - 2012
BEng in Electrical Engineering. Thesis on End-to-end Encryption of Text Messages.

SKILLS

- **Languages:** Python, JavaScript, C++.
- **ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, cuML, Pandas, Polars, Streamlit, Hugging Face, Haystack.
- **Other:** AWS, Git, CI/CD, React, Algorithms and Data Structures.

SELECTED TALKS

- Model Compression with Surrogate Models. AI on Edge for Science and Technology Workshop at the Applied Machine Learning Days (AMLD). February 2025.
- How I Learned to Stop Worrying and Love the Large Language Models: Capabilities, Applications, and Underlying Mechanisms. École Polytechnique Fédérale de Lausanne (EPFL). October 2024.
- AI and Computing Challenges: Harnessing Large Language Models at Thomson Reuters. CERN, KT Seminar. June 2024.
- Optimizing Deep Learning Inference on Resource-constrained Hardware. Université Paris-Saclay, TAU Seminar. June 2023.
- Model Compression for the Hardware Trigger. Rutgers University, HEX Seminar. November 2022.

SELECTED PUBLICATIONS

- Trautmann, D., Ostapuk, N., Grail, Q., Pol, A.A., Bonifazi, G., Gao, S. and Gajek, M., 2024. Measuring the Groundedness of Legal Question-Answering Systems. In Proceedings of the Natural Legal Language Processing Workshop at the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP).
- Pol, A.A., Govorkova, E., Gronroos, S., Chernyavskaya, N., Harris, P., Pierini, M., Ojalvo, I. and Elmer, P., 2023. Knowledge Distillation for Anomaly Detection. In Proceedings of the 31st European Symposium on Artificial Neural Networks (ESANN).
- Zandonati, B., Pol, A.A., Pierini, M., Sirkin, O. and Kopetz, T., 2023. FIT: A Metric for Model Sensitivity. In the Eleventh International Conference on Learning Representations (ICLR).
- Coelho, C.N., Kuusela, A., Li, S., Zhuang, H., Ngadiuba, J., Aarrestad, T.K., Loncar, V., Pierini, M., Pol, A.A. and Summers, S., 2021. Automatic Heterogeneous Quantization of Deep Neural Networks for Low-latency Inference on the Edge for Particle Detectors. Nature Machine Intelligence, 3(8), pp.675-686.
- Pol, A.A., Berger, V., Cerminara, G., Germain, C. and Pierini, M., 2019, December. Trigger Rate Anomaly Detection with Conditional Variational Autoencoders at the CMS Experiment. In Machine Learning and the Physical Sciences Workshop at the 33rd Conference on Neural Information Processing Systems (NeurIPS).
- Pol, A.A., Cerminara, G., Germain, C., Pierini, M. and Seth, A., 2019. Detector Monitoring with Artificial Neural Networks at the CMS Experiment at the CERN Large Hadron Collider. Computing and Software for Big Science, 3, pp.1-13.